

Mathematics of Deep Learning

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Universal approximation theorem

Cybenko's proof

Let $\Omega \subset \mathbb{R}^d$ be compact. Let $\sigma: \mathbb{R} \rightarrow \mathbb{R}$. Let f_θ represents width- N 2-layer neural networks:

$$f_\theta(x) = \sum_{i=1}^N u_i \sigma(a_i^\top x + b_i), \quad (1)$$

where $\theta \in \Theta^{(N)}$ and

$$\Theta^{(N)} = \{(a_1, \dots, a_N, b_1, \dots, b_N, u_1, \dots, u_N) \mid a_1, \dots, a_N \in \mathbb{R}^d, b_1, \dots, b_N, u_1, \dots, u_N \in \mathbb{R}\}.$$

To clarify, $\Theta^{(N)} \cong \mathbb{R}^{dN+2N}$.

Theorem 1. *Let $\Omega \subset \mathbb{R}^d$ be compact. Let $\sigma: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying*

$$\lim_{r \rightarrow -\infty} \sigma(r) = 0, \quad \lim_{r \rightarrow \infty} \sigma(r) = 1.$$

The class of functions

$$\bigcup_{N \in \mathbb{N}} \{f_\theta\}_{\theta \in \Theta^{(N)}}$$

is dense in $(\mathcal{C}(\Omega), \|\cdot\|_\infty)$, i.e.,

$$\text{closure}(\text{span}(\{\sigma(a^\top x + b)\}_{a \in \mathbb{R}^d, b \in \mathbb{R}})) = (\mathcal{C}(\Omega), \|\cdot\|_\infty).$$

The consequence of this theorem is that for any $f_\star \in \mathcal{C}(\Omega)$ and $\varepsilon > 0$, there exists a large enough N and network parameter $\theta \in \Theta^{(N)}$ such that

$$\sup_{x \in \Omega} |f_\theta(x) - f_\star| < \varepsilon.$$

Smaller ε will likely require larger N , but this theorem or its proof will not allow us to make any quantitative claims. Moreover, this is an existence result; it does not tell us how to find N and $\theta \in \Theta^{(N)}$.

Pseudo-proof in dimension 1. Let $d = 1$. Note that

$$\sigma\left(\frac{a}{\delta}x - \frac{b}{\delta}\right) \rightarrow \mathbf{1}_{[b/a, \infty)}$$

as $\delta \rightarrow 0$, i.e., for small δ , we have a smooth approximation of the step function. (Note for $x = \frac{b}{a}$, we do not have convergence. Remember that this is not a real proof.) Then

$$\mathbf{1}_{[t_0, t_1]} \approx \sigma\left(\frac{1}{\delta}x - \frac{t_0}{\delta}\right) - \sigma\left(\frac{1}{\delta}x - \frac{t_1}{\delta}\right)$$

for small $\delta > 0$.

Given a smooth function f_* , find a piecewise constant approximation to it. Then form a smooth approximation of the piecewise constant approximation. $\square$

The actual proof of Theorem 1 will be done in two steps, with the following Lemmas 1 and 2. We say σ is *discriminatory* if

$$\left[\mu \in \mathcal{M}(\Omega) \text{ such that } \int_{\Omega} \sigma(a^\top x + b) d\mu(x) = 0, \forall a \in \mathbb{R}^d, b \in \mathbb{R} \right] \Rightarrow \mu = 0.$$

In functional analysis, one often views an object in a primal and a dual way. In the primal view, we view $\mu \in \mathcal{M}(\Omega)$ as mapping that assigns a “volume” to any measurable set. In the dual view, we instead view the action of $L_\mu: C(\Omega) \rightarrow \mathbb{R}$ defined by

$$L_\mu[f] = \int_{\Omega} f(x) d\mu(x).$$

Under this view, σ is discriminatory if the fact that $L_\mu[\sigma(a^\top \cdot + b)] = 0$ for all a and b implies that $L_\mu[f] = 0$ for all f , i.e., to determine whether $L_\mu = 0$, it is sufficient to check all inputs of the form $\sigma(a^\top \cdot + b)$.

We quickly provide some non-examples. Let $d = 1$ and $\Omega = [-10, 10]$. Then $\sigma(x) = 1$ is not discriminatory since

$$\mu = -\delta_{-1} + \delta_1 \neq 0,$$

where δ_r is the Dirac delta measure centered at $r \in \mathbb{R}$, satisfies

$$\int_{\Omega} \sigma(ax + b) = 0, \quad \forall a \in \mathbb{R}^d, b \in \mathbb{R}.$$

Likewise, $\sigma(x) = x$ is not discriminatory since

$$\mu = \frac{1}{2}\delta_{-1} - \delta_0 + \frac{1}{2}\delta_1 \neq 0$$

satisfies

$$\int_{\Omega} \sigma(ax + b) = 0, \quad \forall a \in \mathbb{R}^d, b \in \mathbb{R}.$$

One can show that all polynomials are not discriminatory.

Lemma 1. *Let $\Omega \subseteq \mathbb{R}^d$. If σ is discriminatory, then $\{f_\theta\}$ is dense in $(\mathcal{C}(\Omega), \|\cdot\|_\infty)$.*

Proof. Let $\mathcal{S} = \text{span}(\{\sigma(a^\top x + b)\}_{a \in \mathbb{R}^d, b \in \mathbb{R}}) \subseteq \mathcal{C}(\Omega)$, and let $\overline{\mathcal{S}}$ be its closure in $(\mathcal{C}(\Omega), \|\cdot\|_\infty)$. Assume for contradiction that $\overline{\mathcal{S}} \neq \mathcal{C}(\Omega)$. Then pick $g \neq 0$, $g \in \mathcal{C}(\Omega) \setminus \overline{\mathcal{S}}$ and define the linear form L on $\overline{\mathcal{S}} \oplus \text{span}(g)$ as

$$L[s + \lambda g] = \lambda, \quad \forall s \in \overline{\mathcal{S}}, \lambda \in \mathbb{R}.$$

This makes L a bounded linear operator¹ such that $L = 0$ on $\overline{\mathcal{S}}$ but $L \neq 0$. Using the Hahn–Banach theorem, we can extend L to $\overline{L}: \mathcal{C}(\Omega) \rightarrow \mathbb{R}$ such that $\overline{L}$ is bounded and linear. Since $\overline{L} \in \mathcal{C}(\Omega)^* \cong \mathcal{M}(\Omega)$, there exists $\mu_{\overline{L}} \in \mathcal{M}(\Omega)$ such that $\overline{L}(h) = \int_{\Omega} h d\mu_{\overline{L}}$. However, $\overline{L} = 0$ on $\overline{\mathcal{S}}$, so

$$\int \sigma(a^\top x + b) d\mu_{\overline{L}}(x) = 0$$

for all a, b and $\mu_{\overline{L}} = 0$ and $\overline{L} = 0$ by the hypothesis. $\overline{L} = 0$ contradicts the construction $L[s + \lambda g] = \lambda$, so we conclude $\overline{\mathcal{S}} = \mathcal{C}(\Omega)$. $\square$

¹In the current setup, $\text{dist}(g, \overline{\mathcal{S}}) > 0$, since $\overline{\mathcal{S}}$ is closed, and $\|s + \lambda g\| = |\lambda| \|(1/\lambda)s + g\| \geq |\lambda| \text{dist}(g, \overline{\mathcal{S}})$. So $|L[s + \lambda g]| = |\lambda| \leq \frac{1}{\text{dist}(g, \overline{\mathcal{S}})} \|s + \lambda g\|$, and L is bounded.

Lemma 2. *A σ satisfying the condition of Theorem 1 is discriminatory.*

Proof. Define

$$\varphi_{a,b} = \sigma(a^\top x + b)$$

and

$$\begin{aligned} H_{a,b} &= \{x \mid a^\top x + b > 0\} && \text{(open half-space defined by } (a,b)) \\ \partial H_{a,b} &= \{x \mid a^\top x + b = 0\} && \text{(boundary of the half space)} \end{aligned}$$

for all $a \in \mathbb{R}^d$ and $b \in \mathbb{R}$. Then

$$\varphi_{\frac{a}{\delta}, \frac{b}{\delta} + t}(x) = \sigma\left(\frac{a^\top x + b}{\delta} + t\right) \xrightarrow{\delta \rightarrow 0} \gamma_t = \begin{cases} 1 & \text{if } x \in H_{a,b} \\ \sigma(t) & \text{if } x \in \partial H_{a,b} \\ 0 & \text{otherwise} \end{cases}$$

pointwise. By the Lebesgue dominated convergence theorem (since σ is bounded)

$$\int_{\Omega} \varphi_{a/\delta, b/\delta + t}(x) d\mu(x) \rightarrow \int_{\Omega} \gamma_t(x) d\mu(x) = \sigma(t)\mu(\partial H_{a,b}) + \mu(H_{a,b}).$$

Since the question is whether σ is discriminatory, consider the scenario where all of these integrals vanish. Then $\sigma(t)\mu(\partial H_{a,b}) + \mu(H_{a,b}) = 0$ for all $t \in \mathbb{R}$, $a \in \mathbb{R}^d$, and $b \in \mathbb{R}$. Since σ is not a constant function, this implies $\mu(\partial H_{a,b}) = \mu(H_{a,b}) = 0$. If this implies that $\mu = 0$, then σ is discriminatory.

We now show $\mu = 0$. For $a \in \mathbb{R}^d$, consider

$$F_{a,\mu}[h] = \int_{\Omega} h(a^\top x) d\mu(x),$$

which is linear. We have

$$F_{a,\mu}[\mathbf{1}_{[-b,\infty)}] = \int_{\Omega} \mathbf{1}_{[-b,\infty)}(a^\top x) d\mu(x) = \mu(\partial H_{a,b}) + \mu(H_{a,b}) = 0.$$

We define *step functions* to be functions of the form

$$\sum_{i=1}^N c_i \mathbf{1}_{[t_i, t_{i+1})}$$

with $N \in \mathbb{N}$, $t_1 < t_2 < \dots < t_N$, and $c_1, \dots, c_N \in \mathbb{R}$. By linearity,

$$F_{a,\mu}[h] = 0$$

for all step functions h . There exists a sequence of step functions $h_1, h_2, \dots$ such that

$$|h_i(x)| \leq |\sin(x)|, \quad h_i(x) \rightarrow \sin(x), \quad \forall x \in \mathbb{R}.$$

By the Lebesgue dominated convergence theorem,

$$\int_{\Omega} \sin(a^\top x) d\mu(x) = 0.$$

We can make the same argument with $\cos(x)$. Combining the two cases, we get

$$\hat{\mu}(a) = \int_{\Omega} e^{ia^\top x} d\mu(x) = 0.$$

Since the Fourier transform of μ is zero, we conclude $\mu = 0$. □

But what if the function we wish to approximate is discontinuous? For example, what if we wish to approximate a function $f_\star: \Omega \rightarrow \{1, \dots, k\}$. While one can approximate continuous functions with discontinuous functions, one cannot approximate continuous functions with discontinuous ones in the $\|\cdot\|_\infty$ -norm.

However, we can have the continuous function approximate the discontinuous function is most of the domain.

Theorem 2 (Lusin's theorem). *Let $\Omega \subseteq \mathbb{R}^d$ be compact. Let $f: \Omega \rightarrow \mathbb{R}$ be a measurable function. For any $\varepsilon > 0$, there exists a continuous function $f_\varepsilon: \Omega \rightarrow \mathbb{R}$ and $\Omega' \subseteq \Omega$ such that $\text{Vol}(\Omega \setminus \Omega') < \varepsilon$ and such that*

$$f(x) = f_\varepsilon(x), \quad \forall x \in \Omega'.$$

(Here, Vol denotes the “volume” defined by the Lebesgue measure.)

Theorem 3. *Consider the setup of Theorem 1. Let $f_\star: \Omega \rightarrow \{1, \dots, k\}$ be a (measurable) decision function. For any $\varepsilon > 0$, there exists an f_θ and $\Omega' \subseteq \Omega$ such that $\text{Vol}(\Omega \setminus \Omega') < \varepsilon$ and*

$$|f_\star(x) - f_\theta(x)| < \varepsilon, \quad \forall x \in \Omega'.$$

Proof. By Lusin's theorem, we find f_ε and Ω' such that $f_\varepsilon = f_\star$ on Ω' . Then we appeal to Theorem 1 to find an f_θ such that $f_\theta \approx f_\varepsilon$ on Ω . Then $f_\theta \approx f_\star$ on Ω' . $\square$